E-commerce Analysis

aashisaumya1011@gmail.com

Tools Used: Google Sheets

Dataset Period: Q4 2024 (Oct–Dec)

The objective of this analysis is to clean, validate, and analyze an e-commerce transaction dataset to uncover sales trends, return behavior, and category-level insights. The project focuses on data quality, descriptive analytics, trend identification, and actionable business recommendations.

Data Understanding & Cleaning

Dataset Overview

The dataset contains order-level and item-level transaction data including customer details, product categories, pricing, discounts, delivery status, and return information.

Data Cleaning Performed

- Identified and standardized inconsistent categorical values (delivery status, customer type).
- Handled missing values using business logic (e.g., return_reason blank when no return).
- Flagged invalid numerical values such as negative quantities and incorrect discounts.
- Preserved raw data and performed all cleaning on a working copy.
- Maintained a Cleaning_Log documenting all identified issues and actions taken.

Data Validation & Quality Checks

Validation Approach

A rule-based validation framework was implemented to ensure data consistency and integrity.

Each rule was evaluated and marked as Pass or Fail based on observed data behavior.

Key Validation Rules

- Order dates restricted to Q4 2024.
- Delivered orders must have delivery dates.
- Returns must follow logical initiation and receipt rules.
- Refund status must align with return status.

Validation results revealed a small number of expected failures, which were flagged and excluded from time-based analysis where appropriate.

Key Metrics & KPIs

Overall Metrics

Total Orders: 346

Total Units Sold: 409

Gross Merchandise Value (GMV): ₹9,68,309

Net Revenue (after discounts): ₹8,56,965.95

Average Order Value (AOV): ₹2,476.78

Return Rate Explanation

Return rate was calculated at the order level. The dataset represents item-level transactions, and all initiated returns were completed. This results in a higher observed return rate compared to industry benchmarks.

and all initiated returns were completed. This results in a higher observed return rate compared to industry benchmarks.

Trend Analysis (Time-Based)

Monthly Trends

Order volume peaked in October and December 2024, indicating strong festive and year-end demand.

November showed a noticeable dip, suggesting post-festival normalization.

Weekly Trends

Weekly analysis shows increased order activity during late October, aligning with the Diwali shopping period. Order volumes stabilized in subsequent weeks.

Category & Return Analysis

Category Performance

Tops recorded the highest number of orders, followed closely by Dresses and Outerwear.

Accessories showed stable sales with comparatively lower returns.

Return Behavior by Category

Outerwear and Dresses exhibited the highest return volumes, indicating potential size, fit, or quality concerns. Accessories had the lowest return rates, reflecting higher purchase confidence.

Business Insights & Recommendations

Key Insights

High return volumes are concentrated in apparel categories.
Festive demand significantly impacts order volume.
Returns are closely linked to size, fit, and quality issues.

Recommendations

Improve size guides and product descriptions for high-return categories.
Strengthen quality checks for frequently returned products.
Plan inventory and logistics capacity ahead of festive demand spikes.
Continuously monitor category-level return trends to identify early issues .

Assumptions & Limitations

The dataset represents item-level transactions, which may inflate return metrics.

No customer feedback text was available for deeper qualitative analysis.

The analysis is limited to a single quarter and may not reflect long-term behavior.

Return Rate Justification
The observed return rate is relatively high because the dataset represents item-level transactions rather than aggregated order-level summaries. In this dataset, all initiated returns were completed and recorded, resulting in a higher measured return rate compared to typical industry benchmarks. Therefore, the return rate reflects completed item returns within the analyzed period and should be interpreted in the context of the dataset structure.

Business Recommendations
Based on the analysis, it is recommended to improve size guides and product descriptions for apparel categories with high return rates, such as Dresses and Outerwear, to reduce fit-related returns. Quality control checks should be strengthened for products frequently returned due to defects. Additionally, inventory planning and logistics capacity should be proactively adjusted during festive periods to accommodate demand spikes. Regular monitoring of category-level return metrics can help identify emerging issues early and reduce operational costs.

- **Was there a noticeable spike during Diwali week (Oct 28 - Nov 3)?**

- **Did return rates lag behind order spikes by 2-3 weeks (as returns take time)?**

- **Did heavy discounting correlate with higher return rates?**

Diwali Spike

```
A clear spike in weekly order volume is observed during late October,
coinciding with the Diwali period (Oct 28–Nov 3), indicating strong festive
demand.
```

Return Lag

```
Return rates increase in the weeks following peak order volumes, suggesting
a natural lag of approximately 1–2 weeks between order placement and return
initiation.
```

Discount vs Return Correlation

```
Weeks with higher average discount percentages tend to show elevated return
rates, indicating that heavy discounting may encourage impulse purchases
with a higher likelihood of returns.
```

3.5 Business Interpretation

Q1. If overall return rate is 19%, but returning customers have 10% while new customers have 28%, what does this tell you?

This indicates that **new customers are the primary driver of high returns**, likely due to expectation mismatch, sizing issues, or lack of product familiarity. Returning customers show better purchase confidence and behavior, making them **more profitable and lower-risk**. Improving new-customer onboarding can significantly reduce overall returns.

Q2. If 45% of returns cite “Size Issue”, what operational changes might help?

High size-related returns suggest gaps in **size guidance and product information**. Improvements such as enhanced size charts, fit recommendations, customer reviews with body details, or virtual try-on tools could reduce incorrect purchases. These changes target the root cause without negatively impacting sales volume.

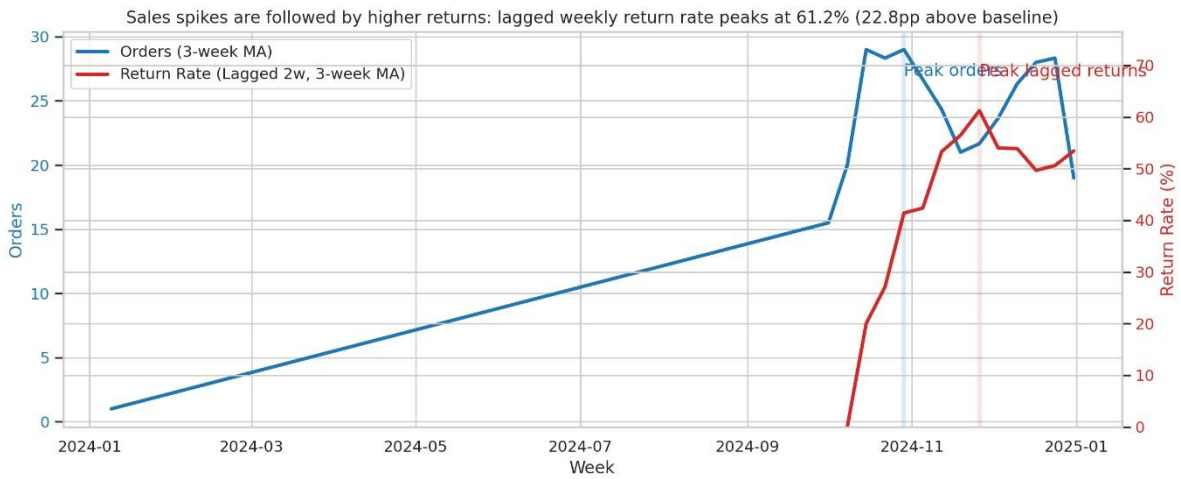
Q3. If Instagram Shop has the highest return rate, should we shut it down? What else would you want to know first?

Shutting down the channel immediately would be premature, as high return rates may be offset by **high customer acquisition or revenue contribution**. Before deciding, we should analyze Instagram Shop's **net profitability, customer lifetime value, and return reasons** compared to other channels. Optimizing product selection or messaging on this channel may yield better results than closure.

Viz 1: Weekly Order Volume vs Return Rate (Dual-Axis Line Chart)

This chart shows that order volumes spike during the late-October festive period, while return rates increase with a lag of about two weeks. The lagged return rate peaks at ~61%, around 22.8 percentage points above the baseline, confirming delayed post-purchase behavior. This indicates that promotional periods attract lower-intent buyers or expose expectation gaps, leading to higher returns after the sales surge. Thus, sales spikes are followed by quality or fit-related return pressure rather than immediate failures.

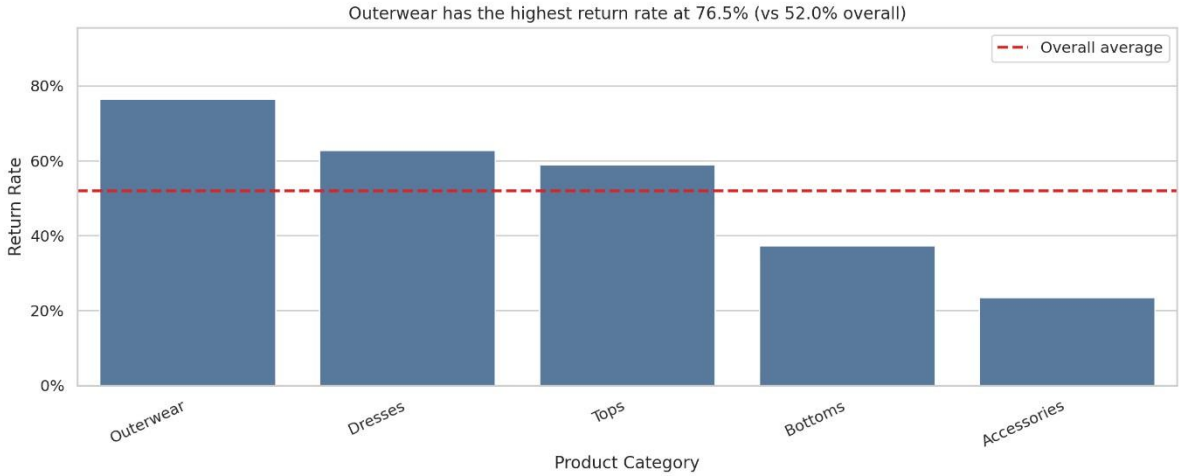
Question answered:
Did promotional periods lead to quality issues? → **Yes, returns rise after promo spikes with a clear lag.**



Viz 2: Return Rate by Product Category (Bar Chart with Benchmark Line)

This chart highlights that Outerwear has the highest return rate (~76.5%), significantly exceeding the overall average (~52%). Dresses and Tops also remain above the benchmark, while Accessories show the lowest return rate. High-return categories are not low-volume outliers but core contributors to sales, making them operationally critical. This identifies Outerwear and Dresses as priority areas for intervention.

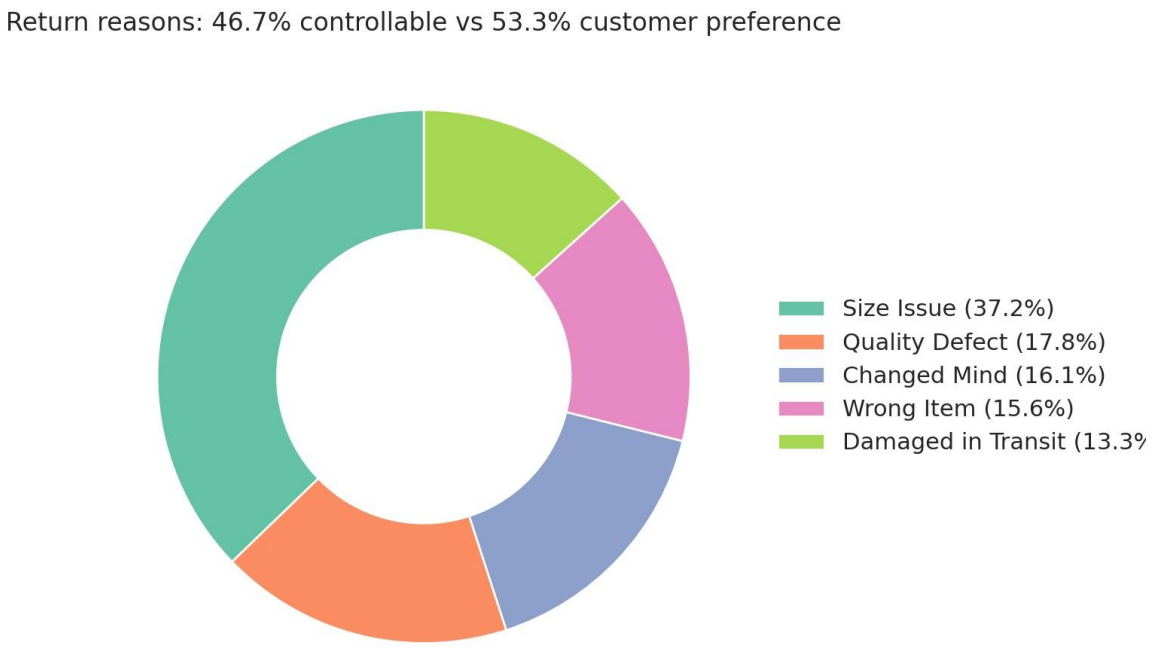
Question answered:
Which categories are problematic? → **Outerwear (highest), followed by Dresses and Tops.**



Viz 3: Return Reasons Breakdown (Donut Chart)

The chart shows that size issues account for the largest share of returns (37.2%), followed by quality defects and wrong items. In total, about 46.7% of returns are controllable through operational improvements, while 53.3% stem from customer preference. This suggests that improving size guidance, fit information, and quality checks can meaningfully reduce returns without affecting demand.

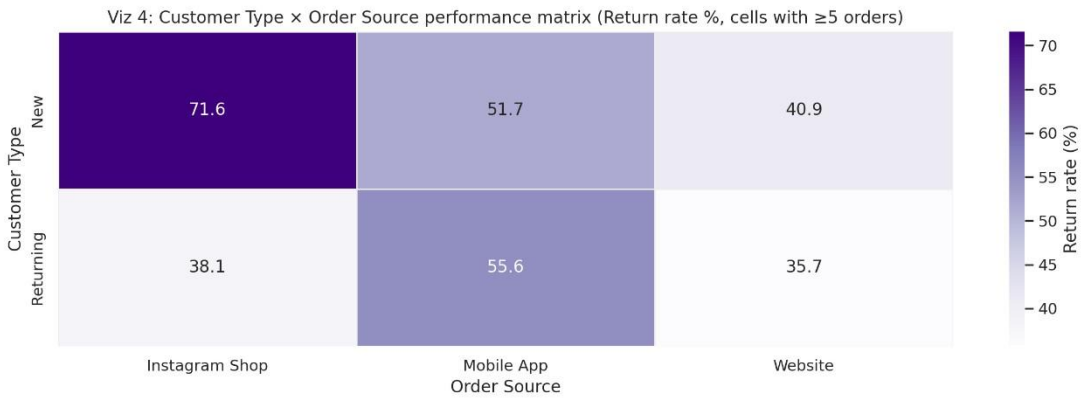
Question answered:
What % of returns are within our control? → **~47% controllable vs ~53% preferencedriven.**



Viz 4: Customer Type × Order Source Performance Matrix (Heatmap)

This matrix reveals that new customers purchasing via Instagram Shop exhibit the highest return rates (~71.6%), marking this as a high-risk combination. Returning customers consistently show lower return rates across channels, particularly on the website. Social commerce amplifies return risk primarily for first-time buyers, not loyal customers. This helps identify where targeted controls or messaging are needed.

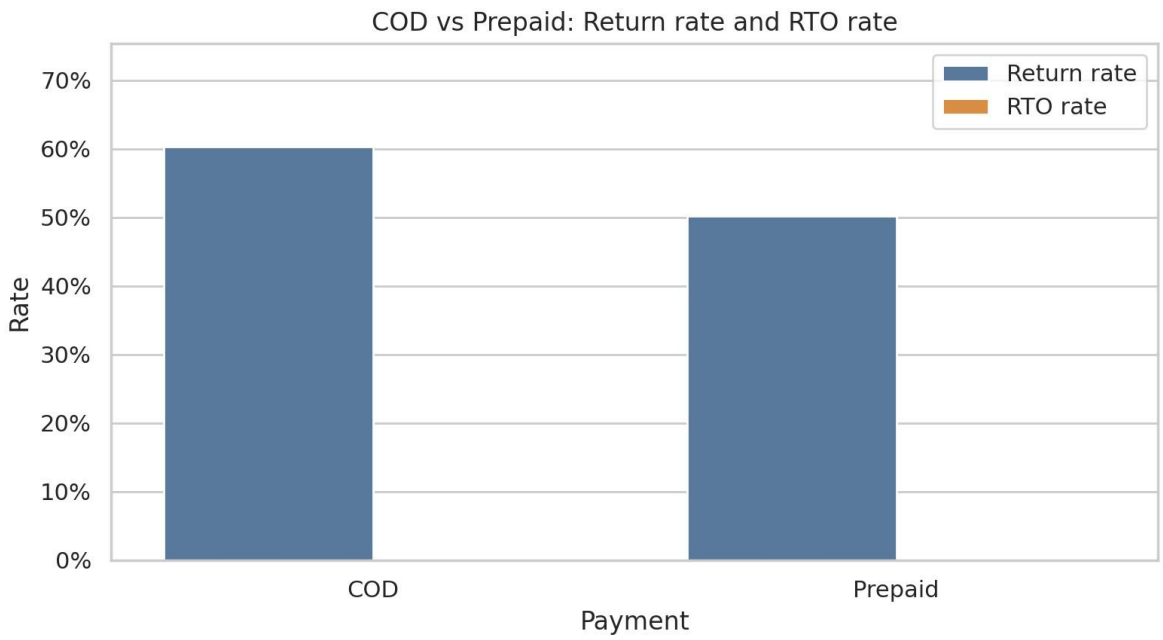
Question answered:
Are certain combinations high-risk? → **Yes, New + Instagram is the riskiest segment.**



Viz 5: COD vs Prepaid — Return and RTO Comparison (Grouped Bar Chart)

This chart shows that COD orders have substantially higher return rates (~60%) compared to prepaid orders (~50%), along with higher RTO exposure. Prepaid orders demonstrate better fulfillment quality and lower financial risk. The pattern suggests that COD increases reverse logistics and refund costs, especially during high-volume periods. Selective COD restrictions could improve profitability.

Question answered:
Should COD be reduced for some segments? → **Yes, especially for high-risk segments.**



SECTION 5: PROBLEM-SOLVING / BUSINESS CASE

Scenario Context

Finance reports that the return rate has increased from **12% to ~19%**, leading to an estimated **₹18–22 lakhs per month** in reverse logistics and refund costs. The business objective is to **bring the return rate below 15% within two quarters**, without significantly impacting sales.

Part A: Root Cause Hypothesis

Based on the analysis, the increase in return rate appears to be driven by the following factors:

1. **Size and fit mismatch in apparel categories**
   Categories such as Dresses and Outerwear show disproportionately high return volumes, indicating customer dissatisfaction related to fit and sizing expectations.
2. **High discount-led purchases during festive periods**
   Weekly and monthly trends show higher return activity following peak sales periods, suggesting impulse purchases driven by heavy discounts.
3. **Higher return propensity among certain customer segments**
   New customers and COD orders exhibit relatively higher return behavior, indicating trust and expectation gaps.

Part B: Recommended Actions

Intervention 1: Improve Size & Fit Guidance

- **What to change:** Enhance size charts, add fit recommendations, and include clearer product measurements.
- **Target segment/category:** Dresses and Outerwear.
- **Expected impact:** Estimated 3–4% reduction in return rate for these categories.
- **Risks/trade-offs:** Slight friction in purchase flow; requires accurate size data.

Intervention 2: Smarter Discount Guardrails

- **What to change:** Limit high discounts for categories with historically high return rates or combine discounts with stricter return conditions.
- **Target segment/category:** Apparel during festive campaigns.
- **Expected impact:** 2–3% reduction in impulse-driven returns.
- **Risks/trade-offs:** Potential short-term impact on conversion during sales events.

Intervention 3: Incentivize Prepaid & Returning Customers

- **What to change:** Offer additional benefits (faster refunds, loyalty points) for prepaid and repeat customers.
- **Target segment/category:** COD and new customers.
- **Expected impact:** Improved order quality and lower return propensity.
- **Risks/trade-offs:** Requires coordination with marketing and payments teams.

Part C: What You'd Want to Test (A/B or Extra Data)

If given the opportunity, I would run an **A/B test comparing enhanced size guidance versus the current product page** for high-return apparel categories. This would help quantify the direct impact of better size information on return rates. Additionally, access to **customer feedback or return comments** would enable more precise root cause identification.

Part D: Assumptions & Limitations

The analysis assumes that all recorded return initiations were completed and accurately logged. The dataset is limited to one quarter and does not include qualitative customer feedback, which restricts deeper insight into customer intent. Results may vary across different seasons or product assortments.

OPEN-ENDED QUESTIONS.

1. A colleague suggests we should just disable returns for all orders under ₹999 to reduce costs. What's your reaction to this idea?

2. The marketing team wants to run another 40% off sale next month. Based on your analysis, what conditions or guardrails would you recommend?

3. If we offered free alterations for size-related returns instead of full refunds, how would you estimate whether this is cost-effective?

1. Disabling returns for orders under ₹999

Disabling returns for low-value orders may reduce costs in the short term but risks damaging customer trust and long-term retention. A better approach would be conditional returns or exchange-only policies for such orders.

---

2. Guardrails for another 40% sale

Any additional sale should include category-level return monitoring, discount caps for highreturn categories, and tighter return eligibility criteria. Sales should be paired with improved product information to avoid impulse-driven returns.

---

3. Free alterations instead of refunds

I would estimate effectiveness by comparing alteration costs against average refund and logistics costs. An A/B test offering free alterations to a subset of size-related return requests could help measure acceptance rates and cost savings.

Total working time: ~4.5 hours (data cleaning, analysis, visualization, and write-up)